system_test

April 1, 2026

[1]:
```python
import numpy as np
from scipy import signal
import tensorflow as tf
from tensorflow.keras import layers, Model, initializers, Sequential
from optic.models.devices import mzm, photodiode, edfa, iqm, coherentReceiver,␣
↪pdmCoherentReceiver, basicLaserModel
from optic.models.channels import linearFiberChannel, ssfm
from optic.comm.modulation import modulateGray, grayMapping
from optic.comm.sources import bitSource, symbolSource
from optic.dsp.core import upsample, pulseShape, pnorm, anorm, signalPower,␣
↪firFilter, decimate, symbolSync,phaseNoise

try:
    from optic.dsp.coreGPU import checkGPU
    if checkGPU():
        from optic.dsp.coreGPU import firFilter
    else:
        from optic.dsp.core import firFilter
except ImportError:
    from optic.dsp.core import firFilter

from optic.utils import parameters, dBm2W, ber2Qfactor
from optic.plot import eyediagram, pconst, plotPSD
import matplotlib.pyplot as plt
from scipy.special import erfc
from tqdm.notebook import tqdm
import scipy as sp
import scipy.constants as const

try:
    from optic.models.modelsGPU import manakovSSF
except:
    from optic.models.channels import manakovSSF

from optic.dsp.equalization import edc, mimoAdaptEqualizer, ffe
from optic.dsp.carrierRecovery import cpr
```

```
from optic.comm.metrics import fastBERcalc, monteCarloGMI, monteCarloMI,␣
 ↪calcEVM, bert
from optic.dsp.clockRecovery import gardnerClockRecovery


import logging as logg
logg.basicConfig(level=logg.INFO, format='%(message)s', force=True)
import time
```

[2]:
```
from IPython.core.display import HTML
from IPython.core.pylabtools import figsize

HTML("""
<style>
.output_png {
    display: table-cell;
    text-align: center;
    vertical-align: middle;
}
</style>
""")
```

[2]: <IPython.core.display.HTML object>

[3]:
```
#␣
 ↪------------------------------------------------------------------------------------------
# Power Amplifier (PA) Functions
#␣
 ↪------------------------------------------------------------------------------------------

def rapp_pa(x, Vsat, p=2.0):
    """
    Memoryless Rapp PA model for complex baseband input.

    Parameters
    ----------
    x : np.ndarray
        Complex input waveform in volts.
    Vsat : float
        Saturation voltage/amplitude.
    p : float
        Rapp smoothness exponent.

    Returns
    -------
    y : np.ndarray
        Output after memoryless PA nonlinearity.
```

```python
    """
    mag = np.abs(x)
    gain = 1.0 / (1.0 + (mag / Vsat) ** (2 * p)) ** (1.0 / (2 * p))
    return x * gain


def butter_lpf_complex(x, Fs, f3dB, order=3):
    """
    Apply an order-N Butterworth LPF to a complex baseband waveform.
    """
    wn = f3dB / (Fs / 2)

    if wn >= 1.0:
        raise ValueError(
            f"Butterworth cutoff must be below Nyquist. Got f3dB={f3dB/1e9:.2f}␣
↪GHz, "
            f"Fs/2={(Fs/2)/1e9:.2f} GHz."
        )

    b, a = signal.butter(order, wn, btype='low')

    y_i = signal.lfilter(b, a, np.real(x))
    y_q = signal.lfilter(b, a, np.imag(x))

    return y_i + 1j * y_q


def pa_model_physical(x, Fs, gain_linear, BLPF_enable, BO_dB=5.0, p=2.0,␣
↪f3dB=10e9, order=3):
    """
    Physical PA model:
        x -> linear gain -> Rapp compression -> Butterworth LPF

    Parameters
    ----------
    x : np.ndarray
        Complex baseband input waveform (dimensionless DSP waveform).
    Fs : float
        Sample rate [Hz].
    gain_linear : float
        Small-signal linear voltage gain. This sets the nominal IQM drive level.
    BO_dB : float
        Back-off in dB, used to set Vsat relative to the RMS value AFTER gain.
    p : float
        Rapp exponent.
    f3dB : float
        PA 3-dB bandwidth [Hz].
```

```
    order : int
        Butterworth filter order.

    Returns
    -------
    y : np.ndarray
        PA output waveform in volts, ready to drive the IQM.
    info : dict
        Diagnostic information.
    """

    # 1) Linear gain stage -> now waveform is in volts
    x_amp = gain_linear * x

    # 2) Compute RMS after gain
    Vrms_in = np.sqrt(np.mean(np.abs(x_amp) ** 2))

    # 3) Saturation voltage from back-off
    Vsat = Vrms_in * 10 ** (BO_dB / 20)

    # 4) Nonlinear compression
    y_nl = rapp_pa(x_amp, Vsat=Vsat, p=p)

    # 5) PA bandwidth limitation
    if BLPF_enable:
        y = butter_lpf_complex(y_nl, Fs=Fs, f3dB=f3dB, order=order)
    else:
        y = y_nl

    info = {
        "gain_linear": gain_linear,
        "Vrms_in_V": Vrms_in,
        "Vsat_V": Vsat,
        "BO_dB": BO_dB,
        "p": p,
        "f3dB_Hz": f3dB,
        "peak_out_V": np.max(np.abs(y)),
        "rms_out_V": np.sqrt(np.mean(np.abs(y) ** 2)),
    }

    return y, info
```

[4]:
```
def select_cpr_mode(paramCPR_, CPR_Mode, Ts, M):
    paramCPR = paramCPR_
    paramCPR.Ts = Ts
    paramCPR.M  = M
    paramCPR.returnPhases = True
```

```python
if CPR_Mode.lower() == "bps":
    paramCPR.alg = "bps"
    if M == 16:
        paramCPR.N = 25
        paramCPR.B = 64
    elif M == 32:
        paramCPR.N = 31
        paramCPR.B = 128
    elif M == 64:
        #paramCPR.N = 81
        paramCPR.N = 101
        paramCPR.B = 2048
        #paramCPR.B = 1024
    elif M == 256:
        paramCPR.N = 81
        paramCPR.B = 512

elif CPR_Mode.lower() == "ddpll":
    paramCPR.alg = "ddpll"
    if M == 16:
        # recommended DDPLL parameters
        paramCPR.tau1 = 1/(2*np.pi*10e3)
        paramCPR.tau2 = 1/(2*np.pi*10e3)
        paramCPR.Kv   = 0.1
    elif M == 32:
        paramCPR.tau1 = 1/(2*np.pi*20e3)
        paramCPR.tau2 = 1/(2*np.pi*20e3)
        paramCPR.Kv   = 0.1

    elif M == 64:
        #paramCPR.tau1 = 1/(2*np.pi*30e3)
        #paramCPR.tau2 = 1/(2*np.pi*30e3)
        #paramCPR.Kv   = 0.15
        paramCPR.Kv = 0.07
        paramCPR.tau1 = 1/(2*np.pi*5e6)
        paramCPR.tau2 = 1/(2*np.pi*5e6)
    elif M == 256:
        paramCPR.tau1 = 1/(2*np.pi*40e3)
        paramCPR.tau2 = 1/(2*np.pi*40e3)
        paramCPR.Kv   = 0.12  # faster tracking needed

else:
    raise ValueError("CPR_Mode must be 'bps' or 'ddpll'")

return paramCPR
```

[5]:
```python
def select_equalizer_mode(M, Data_Aided, paramEq_):
    """
    Selects the equalizer algorithm and step sizes
    based on M, Data_Aided flag, and chosen mode.
    """
    paramEq = paramEq_
    if Data_Aided:
        if M == 4:
            # For QPSK
            paramEq.alg = ['cma', 'cma']
            paramEq.mu  = [5e-3, 1e-3]
            paramEq.numIter = 2
        elif M == 16:
            # For 16-QAM
            paramEq.alg = ['da-rde', 'rde']
            paramEq.mu  = [5e-3, 5e-4]
            paramEq.numIter = 3
        elif M == 32:
            paramEq.alg = ['da-rde', 'rde']
            paramEq.mu  = [2e-3, 5e-4]
            paramEq.numIter = 4
            paramEq.nTaps = 45
        elif M == 64:
            paramEq.alg = ['da-rde', 'rde']
            paramEq.mu  = [8e-4, 4e-4]
            paramEq.mu  = [8e-4, 3e-4]
            paramEq.numIter = 6
            paramEq.nTaps = 65
            #paramEq.alg = ['da-rde', 'cma']
            #paramEq.mu = [3e-4, 5e-5]
            #paramEq.numIter = 4
            #paramEq.nTaps = 85
        elif M == 256:
            paramEq.alg = ['da-rde', 'da-rde']
            paramEq.mu  = [2e-3, 5e-4]
            paramEq.numIter = 6
            paramEq.nTaps = 180


    else:  # Blind mode
        if M == 4:
            paramEq.alg = ['cma','cma']
            paramEq.mu  = [5e-3, 1e-3]
            paramEq.numIter = 2
        elif M == 16:
            paramEq.alg = ['cma','rde']
            paramEq.mu  = [5e-3, 1e-3]
```

```
            paramEq.numIter = 3
        elif M == 32:
            paramEq.alg = ['cma','rde']
            paramEq.mu  = [2e-3, 5e-4]
            paramEq.numIter = 4
            paramEq.nTaps = 45
        elif M == 64:
            paramEq.alg = ['cma', 'rde']
            paramEq.mu  = [1e-3, 1e-3]
            paramEq.numIter = 5
            paramEq.nTaps = 55
        elif M == 256:
            paramEq.alg = ['cma', 'rde']
            paramEq.mu  = [5e-4, 5e-4]
            paramEq.numIter = 6
            paramEq.nTaps = 65


    return paramEq
```

[6]:
```
def perf_calc(symbTx, y_CPR_1, d_, M, paramSymb):
    """Performance Metric Exploration for Single Polarization"""
    d = d_
    discard = 5000
    ind = np.arange(discard, len(symbTx) - discard)

    # Remove phase ambiguity for all M (optional: for QAM)
    if M in [4, 16, 32, 64, 128]:
        d = symbTx  # or processed reference symbols

    # Compute metrics
    BER, SER, SNR = fastBERcalc(y_CPR_1[ind], d[ind], M, 'qam', px=paramSymb.px)
    EVM = calcEVM(y_CPR_1[ind], M, 'qam', d[ind])
    Qfactor = ber2Qfactor(BER[0])

    print(' SER: %.3e,  '%(SER[0]))
    print(' BER: %.3e   '%(BER[0]))
    print(' SNR: %.3f dB'%(SNR[0]))
    print(' EVM: %.3f %%'%(EVM[0]*100))
    print(' Qfactor: %.3f,  '%(Qfactor))

    return BER[0], SER[0], SNR[0], EVM[0], Qfactor
```

[7]:
```
# ----------------------------------------------
# Simulation of the Optical System (parametric version)
# ----------------------------------------------
```

```python
def simulate_optical_system(
    symbTx,
    no_symbols_sent,
    M,
    PA_enable=True,
    Data_Aided=True,
    SpS=16,
    SpSout=2,
    Fs=None,
    mzmScale=0.5,
    Vpi=2,
    BLPF_enable=True,
    PA_BO_dB=3,
    PA_p=2.0,
    PA_f3dB=18.5e9,
    P_launch_dBm=0,
    Rs = 32e9,
    rollOff = 0.01,
    nFilterTaps = 1024,
    laserLinewidth = 100e3,
    FO  = -128e6,
    CPR_Mode = "bps",
    pulse_type="rrc",
    ch_Ltotal_km=80,
    ch_Lspan_km=80,
    ch_alpha_dB_per_km=0.2,
    ch_D_ps_nm_km=16,
    ch_gamma=1.3,
    ch_Fc=193.1e12,
    ch_hz_km=0.5,
    ch_prgsBar=True,
    ch_amp="edfa",
    ch_NF_dB=4.5,
    lo_P_dBm=2,
    lo_RIN_var=0,
    lo_freq_shift_base_hz=0,
    pn_tx_seed=123,
    lo_rx_seed=789,
    pd_seed=1011,
    pd_ideal=True,
    edc_Fs=None,
    pa_order=3,
    ):

    # (derived) params
    if Fs is None:
        Fs = Rs * SpS
```

```python
# 3) UpSampling and FIR Parameters
paramPulse = parameters()
paramPulse.pulseType = pulse_type
paramPulse.nFilterTaps = nFilterTaps
paramPulse.rollOff = rollOff
paramPulse.SpS = SpS

# 4) IQM Parameters
paramIQM = parameters()
paramIQM.Vpi = Vpi
paramIQM.VbI = -Vpi
paramIQM.VbQ = -Vpi
paramIQM.Vphi = Vpi/2

# 5) Optical Carrier / LO field (Ein)
sigTx_length = no_symbols_sent * SpS
if laserLinewidth and laserLinewidth > 0:
    phi_pn = phaseNoise(laserLinewidth, sigTx_length, 1 / Fs,␣
↪seed=pn_tx_seed)
    sigLO = np.exp(1j * phi_pn)
else:
    sigLO = np.ones_like(sigTx_length, dtype=complex)


# ------------------------------------------------
# Channel Parameters
#------------------------------------------------

# 1) Optical Channel Parameters
paramCh = parameters()
paramCh.Ltotal = ch_Ltotal_km
paramCh.Lspan = ch_Lspan_km
paramCh.alpha = ch_alpha_dB_per_km
paramCh.D = ch_D_ps_nm_km
paramCh.gamma = ch_gamma
paramCh.Fc = ch_Fc
paramCh.hz = ch_hz_km
paramCh.prgsBar = ch_prgsBar
paramCh.Fs = Fs
paramCh.amp = ch_amp
paramCh.NF = ch_NF_dB
#paramCh.seed = 456


# ------------------------------------------------
# Receiver Parameters
#------------------------------------------------
```

```python
# 1) local oscillator (LO) parameters:

paramLO = parameters()
paramLO.P = lo_P_dBm
paramLO.lw = laserLinewidth
paramLO.RIN_var = lo_RIN_var
paramLO.Fs = Fs
paramLO.seed = lo_rx_seed
paramLO.freqShift = lo_freq_shift_base_hz + FO

# 2) Front-End Parameters and photodiode paramters

# Frontend parameters
paramFE = parameters()
paramFE.Fs = Fs

# Photodiodes parameters
paramPD = parameters()
paramPD.B = Rs
paramPD.Fs = Fs
paramPD.ideal = pd_ideal
paramPD.seed = pd_seed

# 3) Pulseshaping in the reciever using rrc filter
paramRxPulse = parameters()
paramRxPulse.SpS = SpS
paramRxPulse.nFilterTaps = nFilterTaps
paramRxPulse.rollOff = rollOff
paramRxPulse.pulseType = pulse_type

# 4) Decimation Parameters
paramDec = parameters()
paramDec.SpSin  = SpS
paramDec.SpSout = SpSout

# 5) Chromatic Dispersion Parameters
paramEDC = parameters()
paramEDC.L = paramCh.Ltotal
paramEDC.D = paramCh.D
paramEDC.Fc = paramCh.Fc
paramEDC.Rs = Rs
paramEDC.Fs = 2 * Rs if edc_Fs is None else edc_Fs

# 6) Adaptive Equalization Parameters
paramEq = parameters()
paramEq.nTaps = 35
```

```
    paramEq.SpS = paramDec.SpSout
    paramEq.numIter = 2
    paramEq.storeCoeff = False
    paramEq.M = M
    paramEq.shapingFactor = 0
    paramEq.constType = "qam"
    paramEq.prgsBar = False

    # 7) Data-Aided or Blind Reciever Equalization
    # Can be set here or not
    #Data_Aided = True

    # 8) Carrier and Phase recovery parameters using bps
    paramCPR = parameters()
    paramCPR.alg = 'bps'
    paramCPR.M   = M
    paramCPR.constType ="qam"
    paramCPR.shapingFactor = 0
    paramCPR.N   = 25
    paramCPR.B   = 64
    paramCPR.returnPhases = True
    paramCPR.Ts = 1/Rs




    # ------------------------------------------------------------------------------------------
    # TRANSMITTER

    # 2) Upsampling + pulse shaping
    pulse = pulseShape(paramPulse)
    symbolsUp = upsample(symbTx, SpS)
    sigTx = firFilter(pulse, symbolsUp)

    # 3) Choose nominal small-signal PA gain so the nominal drive is around mzmScale * Vpi
    target_peak_V = mzmScale * Vpi
    peak_sigTx = np.max(np.abs(sigTx))

    if peak_sigTx == 0:
        raise ValueError("sigTx peak is zero; cannot set PA gain.")

    gain_linear = target_peak_V / peak_sigTx

    # 4) Driver amplifier / PA output directly in volts
    if PA_enable:
```

```python
        u_drive, paInfo = pa_model_physical(
            sigTx,
            Fs=Fs,
            gain_linear=gain_linear,
            BLPF_enable=BLPF_enable,
            BO_dB=PA_BO_dB,
            p=PA_p,
            f3dB=PA_f3dB,
            order=pa_order,
        )
    else:
        u_drive = gain_linear * sigTx
        paInfo = {
            "gain_linear": gain_linear,
            "Vrms_in_V": np.sqrt(np.mean(np.abs(u_drive) ** 2)),
            "Vsat_V": None,
            "BO_dB": None,
            "p": None,
            "f3dB_Hz": None,
            "peak_out_V": np.max(np.abs(u_drive)),
            "rms_out_V": np.sqrt(np.mean(np.abs(u_drive) ** 2)),
        }

    # 5) IQ modulation: PA output drives the IQM directly
    sigTxo = iqm(sigLO, u_drive, paramIQM)

    # 6) Set launched optical power
    P_launch_W = dBm2W(P_launch_dBm)
    sigTxo = np.sqrt(P_launch_W) * pnorm(sigTxo)

    # End of Transmitter
    # ------------------------------------------------------------------------------------------

    # ------------------------------------------------------------------------------------------
    # CHANNEL

    sigCh = ssfm(sigTxo, paramCh)

    # End of CHANNEL
    # ------------------------------------------------------------------------------------------

    # ------------------------------------------------------------------------------------------
    # RECEIVER
```

```python
# 1) Generate CW laser LO field
paramLO.Ns = len(sigCh)
sigLO_Rx = basicLaserModel(paramLO)

# 2) Coherent receiver for single-polarization
sigRxFrontEnd = coherentReceiver(sigCh, sigLO_Rx, paramFE, paramPD)

# 3) Pulse shaping
pulse = pulseShape(paramRxPulse)
sigRxPulseShape = firFilter(pulse, sigRxFrontEnd)

# 4) Decimation
sigRxDecimation = decimate(sigRxPulseShape, paramDec)

# 5) Chromatic Dispersion Compensation
sigRxCD = edc(sigRxDecimation, paramEDC)

# 6) Symbol Synchronization with the SymbTx
symbRxCD = symbolSync(sigRxCD, symbTx, 2)

# 7) Power Normalization
x = pnorm(sigRxCD)
d = pnorm(symbRxCD)

if M==256 and Data_Aided:
    paramEq.L = [int(0.5*d.shape[0]), int(0.5*d.shape[0])]
else:
    paramEq.L = [int(0.2*d.shape[0]), int(0.8*d.shape[0])]

#paramEq.L         = [int(0.8 * d.shape[0])]    # or d.shape[0] - 20k
# -------------------------------------------------
# 5) EQUALIZATION (via DSP SWITCH)
# -------------------------------------------------
paramEq = select_equalizer_mode(M, Data_Aided, paramEq)

if Data_Aided:
    print(" adied")
    y_EQ = mimoAdaptEqualizer(x, paramEq, d)
else:
    print("no adied")
    y_EQ = mimoAdaptEqualizer(x, paramEq, None)

#y_EQ, h_rls = rls_single_pol(x, d, L=21, lam=0.995, delta=1e3)

# -------------------------------------------------
# 6) Frequency Offset Compensation
```

```
    # ------------------------------------------------
    Ts = 1 / Rs
    paramCPR = select_cpr_mode(paramCPR, CPR_Mode, Ts, M)
    if CPR_Mode == "ddpll":
        print("no bps")

        y_EQ_2D = y_EQ.reshape(-1,1) if y_EQ.ndim == 1 else y_EQ

        symbTx_2D = symbTx.reshape(-1,1)
        y_CPR_1, phaseEst = cpr(y_EQ_2D, param=paramCPR, symbTx=symbTx_2D)
        y_CPR_1= y_CPR_1.flatten()
    else:
        print("yes bps")
        y_CPR_1, phaseEst = cpr(y_EQ, paramCPR)

    return y_CPR_1, d, phaseEst
```

[8]:
```
def intialise_paramSymb(M, nBits, seed=444):
    # Symbol generation
    paramSymb = parameters()
    paramSymb.nSymbols = int(nBits // np.log2(M))  # symbols = bits / log2(M)
    paramSymb.M = M
    paramSymb.constType = "qam"                    # 'qam' with M=4 -> QPSK
    paramSymb.dist = "uniform"                     # uniform symbol␣
 ↪probabilities
    paramSymb.seed = 444
    paramSymb.shapingFactor = 0

    constSymb = grayMapping(paramSymb.M, paramSymb.constType)
    if paramSymb.dist == "uniform":
        px = np.ones(paramSymb.M) / paramSymb.M
    elif paramSymb.probDist == "maxwell-boltzmann":
        px = np.exp(-paramSymb.shapingFactor * np.abs(constSymb) ** 2)
        px = px / np.sum(px)
    else:
        raise ValueError("Invalid probability distribution.")
    paramSymb.px = px
    return paramSymb
```

0.1 System First Trial - No DPD

[9]:
```
# M = 16
# nBits = 400000
# SpSout = 2
# paramSymb = intialise_paramSymb(M, nBits)
```

[10]:
```
# symbTx = symbolSource(paramSymb)



# y_CPR_1, d, phaseEst = simulate_optical_system(symbTx, len(symbTx), M,␣
↪PA_enable=True, Data_Aided=True, SpSout=SpSout, mzmScale=0.8)



# perf_calc(symbTx, y_CPR_1, d, M, paramSymb)



# discard = 5000
# # plot constellations
# pconst(y_CPR_1[discard:-discard])

# # plotting eye diagrams of sigTx
# eyediagram(y_CPR_1.real[discard:-discard], y_CPR_1.real.size-2*discard,␣
↪SpSout, plotlabel='signal at Rx REAL', ptype='fancy')
# eyediagram(y_CPR_1.imag[discard:-discard], y_CPR_1.imag.size-2*discard,␣
↪SpSout, plotlabel='signal at Rx IMAGNIARY', ptype='fancy')
```

0.2 System Benchmark (No DPD)

[11]:
```
# results = []

# # modulation orders to test
# mod_orders = [16,64,256]

# # DA and CPR modes
# modes_DataAided = [True, False]
# modes_CPR = [ "bps","ddpll"]

# for M_test in mod_orders:
#     M = M_test
#     paramSymb = intialise_paramSymb(M, nBits) # need to update it everytime M␣
↪changes
#     for da in modes_DataAided:
#         for cpr_mode in modes_CPR:
#             print("\n=====================================================")
#             print(f" Running:  M={M}, Data_Aided={da}, CPR_Mode={cpr_mode}")
#             print("=====================================================\n")

#             # fresh symbol stream each run
#             symbTx = symbolSource(paramSymb)

#             # run sys
#             y_CPR_1, d, phaseEst = simulate_optical_system(symbTx,␣
↪len(symbTx), M,
```

```
#                                                               Data_Aided=da,
#                                                               SpSout=SpSout,
#                                                               CPR_Mode=cpr_mode)

#             discard = 5000

#             # optional: plot constellations
#             pconst(y_CPR_1[discard:-discard])

#             # plotting eye diagrams
#             eyediagram(y_CPR_1.real[discard:-discard],
#                         y_CPR_1.real.size-2*discard,
#                         SpSout,
#                         plotlabel=f'signal at Rx REAL, M={M}', ptype='fancy')
#             eyediagram(y_CPR_1.imag[discard:-discard],
#                         y_CPR_1.imag.size-2*discard,
#                         SpSout,
#                         plotlabel=f'signal at Rx IMAGINARY, M={M}',␣
↪ptype='fancy')


#             # performance
#             BER, SER, SNR, EVM, Q = perf_calc(symbTx, y_CPR_1, d, M,␣
↪paramSymb)


#             results.append({
#                 "Modulation": M,
#                 "Data_Aided": da,
#                 "CPR": cpr_mode,
#                 "BER": BER,
#                 "SER": SER,
#                 "SNR": SNR,
#                 "EVM": EVM,
#                 "Qfactor": Q
#             })
```

[12]:
```
# print("\n==================== SUMMARY TABLE ====================\n")
# for r in results:
#     print(f"DA={r['Data_Aided']}, CPR={r['CPR']}: "
#           f"BER={r['BER']:.2e}, SER={r['SER']:.2e}, "
#           f"SNR={r['SNR']:.2f} dB, EVM={r['EVM']*100:.1f} %, Q={r['Qfactor']:.
↪2f}")
```

1 DPD

[13]:
```
def build_model():
    inputs = layers.Input(shape=(None, 2)) # 2 for I and Q

    # sec_a = layers.DepthwiseConv1D(100, padding='same',␣
 ↪use_bias=False)(inputs) # 100 taps was a sweet spot, 20-ish fails to␣
 ↪converge, tiker with different values.

    # 1. Separate the I and Q channels
    # Using Lambda layers is often cleaner for serialization than tf.split
    input_i = layers.Lambda(lambda x: x[:, :, 0:1])(inputs)
    input_q = layers.Lambda(lambda x: x[:, :, 1:2])(inputs)

    # 2. Causal FIR filters (No bias to match your HLS 'acc' logic)
    # Name them so you can extract weights easily by name!
    sec_a_i = layers.Conv1D(1, 100, padding='causal', use_bias=False,␣
 ↪name='fir_i')(input_i)
    sec_a_q = layers.Conv1D(1, 100, padding='causal', use_bias=False,␣
 ↪name='fir_q')(input_q)

    # 3. Re-combine for the Dense layers
    sec_a = layers.Concatenate(axis=-1)([sec_a_i, sec_a_q])


#    sec_a = inputs

    # nonlinear_1 = layers.Dense(20, activation=layers.LeakyReLU(0.1))(sec_a)
    # nonlinear_2 = layers.Dense(20, activation=layers.LeakyReLU(0.
 ↪1))(nonlinear_1)
    # nonlinear_3 = layers.Dense(2, activation=layers.LeakyReLU(0.
 ↪1))(nonlinear_2)
    # nonlinear_4 = layers.Dense(2, activation='linear')(nonlinear_3)


    nonlinear_1 = layers.Dense(20, activation=tf.math.sin)(sec_a)
    nonlinear_2 = layers.Dense(20, activation=tf.math.sin)(nonlinear_1)
    # nonlinear_3 = layers.Dense(2, activation=tf.math.sin)(nonlinear_2)
    nonlinear_3 = layers.Dense(2, activation='linear')(nonlinear_2)


    # nonlinear_1 = layers.Dense(20, activation='tanh')(sec_a)
    # nonlinear_2 = layers.Dense(20, activation='tanh')(nonlinear_1)
    # nonlinear_3 = layers.Dense(2, activation='tanh')(nonlinear_2)
    # nonlinear_4 = layers.Dense(2, activation='linear')(nonlinear_3)
```

```
    outputs = layers.Add()([sec_a, nonlinear_3])

    # outputs = nonlinear_4
    model = Model(inputs, outputs)
    model.compile(optimizer=tf.keras.optimizers.Adam(learning_rate=1e-2),␣
 ↪loss='mse') # ILA)
    return model
```

[14]:
```
def split_i_q(arr):
    return np.stack([np.real(arr), np.imag(arr)]).T

def merge_i_q(arr):
    if arr.ndim == 3:
        return arr[:,:,0] + 1j*arr[:,:,1]
    elif arr.ndim == 2:
        return arr[:,0] + 1j*arr[:,1]
    else:
        raise ValueError("Input array must be 2D or 3D.")

def preprocess(symbTx, seq_length=5000):
    reshabe_len = len(symbTx)//seq_length
    symbTx_nn = split_i_q(symbTx)
    symbTx_nn = symbTx_nn[:reshabe_len*seq_length].reshape(-1, seq_length, 2) #␣
 ↪batching the symbols for training (shape: num_batches, seq_length,␣
 ↪num_features)
    return symbTx_nn

def postprocess(symbTx_nn, original_symbTx, seq_length=5000):
    reconstructed_nn = symbTx_nn.reshape(-1, 2)
    reconstructed_complex = reconstructed_nn[:, 0] + 1j * reconstructed_nn[:, 1]

    total_len = len(original_symbTx)
    cutoff_point = (total_len // seq_length) * seq_length
    thrown_off_symbols = original_symbTx[cutoff_point:]

    return np.concatenate([reconstructed_complex, thrown_off_symbols])
```

[15]:
```
def train_DPD(M, nBits, iteration_cnt = 15, **kwargs):

    paramSymb = intialise_paramSymb(M, nBits, seed=333)
    symbTx = symbolSource(paramSymb)
    dpd_model = build_model()
    symbTx_nn = preprocess(symbTx)
    dpd_model.fit(symbTx_nn, symbTx_nn, epochs=500, verbose=0) # this line is␣
 ↪important, = starting as a passthrough.
```

```
    best_ber = float('inf')

    for iteration in range(iteration_cnt):
        print(f"====== Iteration {iteration} ======")

        symbDPD = dpd_model.predict(symbTx_nn, verbose=0)
        x = merge_i_q(symbDPD).flatten()
        # x =   postprocess(symbDPD, symbTx)

        y_CPR_1, d, phaseEst = simulate_optical_system(x, len(x), M, **kwargs)


        ber, SER, SNR, EVM, Q = perf_calc(symbTx, y_CPR_1, d, M, paramSymb)
        if ber < best_ber:
            dpd_model.save_weights("best_model.weights.h5")
            best_ber = ber


        y_CPR_1_nn = preprocess(y_CPR_1)


        # y = (y_CPR_1 - np.mean(y_CPR_1)) / np.std(y_CPR_1)
        # y_nn = split_i_q(y).reshape(-1, seq_length, 2)

        dpd_model.fit(y_CPR_1_nn, symbDPD, epochs=100, verbose=0) # ILA


        ## PLEASE IGNORE THE BELOW CODE
        # aux_model.fit(symbDPD, y_CPR_1_nn, epochs=200, verbose=0)
        # aux_model.trainable = False
        # dla_cascade.compile(optimizer=tf.keras.optimizers.
↪Adam(learning_rate=1e-3), loss='mse')

        # dla_cascade.fit(symbTx_nn, symbTx_nn, epochs=100, verbose=0)
        # aux_model.trainable = True
        # dla_cascade.compile(optimizer=tf.keras.optimizers.
↪Adam(learning_rate=1e-3), loss='mse')
```

2 Benchmarking W/ DPD

2.0.1 Training the DPD - change to your modulation scheme of choice and re-run this cell

[16]:
```
M = 16
no_symbols= 100_000 # must be multiple of 5000 (seq_length)
nBits = int(no_symbols * np.log2(M))
SpSout = 2
```

```python
mzmScale = 0.8
laserLinewidth = 100e3

dpd_model = build_model()
train_DPD(M, nBits, iteration_cnt=1, Data_Aided=True,
    SpSout=SpSout,mzmScale=mzmScale, CPR_Mode="bps",
    laserLinewidth=laserLinewidth) # here you can specify things like
    back_to_back enable, CPR mode, ..... etc but DONT change Data_Aided - for
    training this must be true


dpd_model.load_weights("best_model.weights.h5")
```

```
2026-04-01 20:54:15.935395: I metal_plugin/src/device/metal_device.cc:1154]
Metal device set to: Apple M3
2026-04-01 20:54:15.935423: I metal_plugin/src/device/metal_device.cc:296]
systemMemory: 24.00 GB
2026-04-01 20:54:15.935428: I metal_plugin/src/device/metal_device.cc:313]
maxCacheSize: 8.88 GB
2026-04-01 20:54:15.935443: I
tensorflow/core/common_runtime/pluggable_device/pluggable_device_factory.cc:305]
Could not identify NUMA node of platform GPU ID 0, defaulting to 0. Your kernel
may not have been built with NUMA support.
2026-04-01 20:54:15.935452: I
tensorflow/core/common_runtime/pluggable_device/pluggable_device_factory.cc:271]
Created TensorFlow device (/job:localhost/replica:0/task:0/device:GPU:0 with 0
MB memory) -> physical PluggableDevice (device: 0, name: METAL, pci bus id:
<undefined>)
2026-04-01 20:54:16.306070: I
tensorflow/core/grappler/optimizers/custom_graph_optimizer_registry.cc:117]
Plugin optimizer for device_type GPU is enabled.
```

```
====== Iteration 0 ======
```

```
  0%|          | 0/1 [00:00<?, ?it/s]
```

```
Running CD compensation…
CD filter length: 46 taps, FFT size: 64
Running adaptive equalizer…
da-rde - training stage #0
da-rde pre-convergence training iteration #0
```

```
 adied
```

```
da-rde MSE = 0.041518.
da-rde pre-convergence training iteration #1
da-rde MSE = 0.035654.
da-rde pre-convergence training iteration #2
da-rde MSE = 0.035573.
rde - training stage #1
```

```
rde MSE = 0.024337.
Running frequency offset compensation…
Estimated frequency offset (MHz): [128.24]
Running BPS carrier phase recovery…

yes bps

Estimated linewidth: 388.021 kHz

 SER: 8.333e-04,
 BER: 2.083e-04
 SNR: 19.381 dB
 EVM: 1.162 %
 Qfactor: 5.477,

/Users/abednaser/major_proj/.venv/lib/python3.12/site-
packages/keras/src/saving/saving_lib.py:797: UserWarning: Skipping variable
loading for optimizer 'adam', because it has 2 variables whereas the saved
optimizer has 18 variables.
  saveable.load_own_variables(weights_store.get(inner_path))
```

2.0.2 Performance Evaluation / Testing

[17]:
```python
# # TODO: validate the accuracy of the postprocess func.

# results_nodpd = []
# results_dpd = []

# # DA and CPR modes
# modes_DataAided = [True, False]
# modes_CPR = [ "bps"]

# paramSymb = intialise_paramSymb(M, nBits, seed=333)

# for da in modes_DataAided:
#     for cpr_mode in modes_CPR:
#             print("\n=====================================================")
#             print(f" Running:  M={M}, Data_Aided={da}, CPR_Mode={cpr_mode}")
#             print("=====================================================\n")

#             # fresh symbol stream each run, ... do i need to change the seed?
#             symbTx = symbolSource(paramSymb)


#             ###################### W/O DPD ######################
#             y_CPR_1, d, phaseEst = simulate_optical_system(symbTx,␣
↪len(symbTx), M,
#                                                            Data_Aided=da,
#                                                            SpSout=SpSout,
```

```
#                                                                          ␣
↪CPR_Mode=cpr_mode, mzmScale=mzmScale, laserLinewidth=laserLinewidth)

#              BER, SER, SNR, EVM, Q = perf_calc(symbTx, y_CPR_1, d, M,␣
↪paramSymb)

#              results_nodpd.append({
#                  "Modulation": M,
#                  "Data_Aided": da,
#                  "CPR": cpr_mode,
#                  "BER": BER,
#                  "SER": SER,
#                  "SNR": SNR,
#                  "EVM": EVM,
#                  "Qfactor": Q
#              })

#              ###################### W/ DPD ######################

#              symbTx_nn = preprocess(symbTx)
#              symbDPD = dpd_model.predict(symbTx_nn, verbose=0)
#              symbDPD = merge_i_q(symbDPD).flatten()

#              # run sys
#              y_CPR_1, d, phaseEst = simulate_optical_system(symbDPD,␣
↪len(symbDPD), M,
#                                                             Data_Aided=da,
#                                                             SpSout=SpSout,
#                                                                          ␣
↪CPR_Mode=cpr_mode, mzmScale=mzmScale, laserLinewidth=laserLinewidth
#                                                             )

#              # performance
#              BER, SER, SNR, EVM, Q = perf_calc(symbTx, y_CPR_1, d, M,␣
↪paramSymb)


#              results_dpd.append({
#                  "Modulation": M,
#                  "Data_Aided": da,
#                  "CPR": cpr_mode,
#                  "BER": BER,
#                  "SER": SER,
#                  "SNR": SNR,
#                  "EVM": EVM,
#                  "Qfactor": Q
#              })
```

[18]:
```
# print("\n==================== W/O DPD Results ====================\n")
# for r in results_nodpd:
#     print(f"DA={r['Data_Aided']}, CPR={r['CPR']}: "
#           f"BER={r['BER']:.2e}, SER={r['SER']:.2e}, "
#           f"SNR={r['SNR']:.2f} dB, EVM={r['EVM']*100:.1f} %, Q={r['Qfactor']:.
↪2f}")

# print("\n==================== W/ DPD Results ====================\n")
# for r in results_dpd:
#     print(f"DA={r['Data_Aided']}, CPR={r['CPR']}: "
#           f"BER={r['BER']:.2e}, SER={r['SER']:.2e}, "
#           f"SNR={r['SNR']:.2f} dB, EVM={r['EVM']*100:.1f} %, Q={r['Qfactor']:.
↪2f}")
```

[19]:
```
dpd_model.summary()
```

Model: "functional"

Layer (type)	Output Shape	Param #	Connected to
input_layer (InputLayer)	(None, None, 2)	0	-
lambda (Lambda)	(None, None, 1)	0	input_layer[0][0]
lambda_1 (Lambda)	(None, None, 1)	0	input_layer[0][0]
fir_i (Conv1D)	(None, None, 1)	100	lambda[0][0]
fir_q (Conv1D)	(None, None, 1)	100	lambda_1[0][0]
concatenate (Concatenate)	(None, None, 2)	0	fir_i[0][0], fir_q[0][0]
dense (Dense)	(None, None, 20)	60	concatenate[0][0]
dense_1 (Dense)	(None, None, 20)	420	dense[0][0]
dense_2 (Dense)	(None, None, 2)	42	dense_1[0][0]
add (Add)	(None, None, 2)	0	concatenate[0][0… dense_2[0][0]

Total params: 722 (2.82 KB)

Trainable params: 722 (2.82 KB)

Non-trainable params: 0 (0.00 B)

[31]:

```
fir_i = dpd_model.get_layer('fir_i')
print("const float taps_I[NO_TAPS] = {")
for tap in fir_i.get_weights()[0].reshape(-1):
    print(f"    {tap},")
print("};")

fir_q = dpd_model.get_layer('fir_q')
print("const float taps_Q[NO_TAPS] = {")
for tap in fir_q.get_weights()[0].reshape(-1):
    print(f"    {tap},")
print("};")

#################
fcnn_1  = dpd_model.get_layer('dense')


print("const float weights_1[NEURONS_1*2] = {")
for weight in fcnn_1.get_weights()[0].reshape(-1):
    print(f"    {weight},")
print("};")

print("const float biases_1[NEURONS_1] = {")
for bias in fcnn_1.get_weights()[1].reshape(-1):
    print(f"    {bias},")
print("};")

################
fcnn_2  = dpd_model.get_layer('dense_1')


print("const float weights_2[NEURONS_2*NEURONS_1] = {")
for weight in fcnn_2.get_weights()[0].reshape(-1):
    print(f"    {weight},")
print("};")

print("const float biases_2[NEURONS_2] = {")
for bias in fcnn_2.get_weights()[1].reshape(-1):
    print(f"    {bias},")
print("};")

################
fcnn_3  = dpd_model.get_layer('dense_2')
```

```python
print("const float weights_3[NEURONS_3*NEURONS_2] = {")
for weight in fcnn_3.get_weights()[0].reshape(-1):
    print(f"    {weight},")
print("};")

print("const float biases_3[NEURONS_3] = {")
for bias in fcnn_3.get_weights()[1].reshape(-1):
    print(f"    {bias},")
print("};")
```

```
const float taps_I[NO_TAPS] = {
    -3.3820190310507314e-06,
    -1.0238727554678917e-05,
    -6.683487299596891e-06,
    -7.028203071968164e-06,
    -4.966149390384089e-06,
    6.577192380063934e-06,
    2.9562527288362617e-06,
    3.7120225897524506e-06,
    -7.164100225054426e-06,
    -5.001861609343905e-06,
    6.2049730331636965e-06,
    -3.260575567765045e-06,
    4.753218490805011e-06,
    5.745338512497256e-06,
    3.0023336421436397e-06,
    -1.050653554557357e-05,
    -1.1163312052531182e-08,
    -2.9429886581056053e-06,
    -1.404973772878293e-05,
    8.937915481510572e-06,
    1.018260263663251e-06,
    1.588466329849325e-05,
    -1.3499654869519873e-06,
    1.863744910224341e-05,
    1.08027006717748e-06,
    -2.3989848614291986e-06,
    -1.3309527275850996e-06,
    -4.9110431064036675e-06,
    -1.1376082511560526e-05,
    3.584353862606804e-07,
    -3.255871888541151e-06,
    2.644549340402591e-06,
    2.4228429538197815e-06,
    7.202749657153618e-06,
    -6.238335572561482e-06,
```

```
2.89124318442191e-06,
2.1910032046434935e-06,
-2.72181546279171e-06,
5.5937857723620255e-06,
1.4067354641156271e-06,
8.355589670827612e-06,
4.9557943384570535e-06,
8.41720338939922e-06,
-9.985423275793437e-06,
-9.959497219824698e-06,
2.8258859856578056e-06,
-9.681170922704041e-06,
2.003186409638147e-06,
-3.1599420253769495e-06,
-2.148322664652369e-06,
4.11686914958409e-06,
-1.6433424434580957e-06,
-8.168650310835801e-06,
5.663689989887644e-06,
-9.56673284235876e-06,
-2.0956151729478734e-06,
8.601834906585282e-07,
3.0133037398627494e-06,
-2.8755173389072297e-06,
4.634330252883956e-06,
7.4673575909400824e-06,
-1.836850174186111e-06,
-3.504028427414596e-06,
-9.666482583270408e-07,
-4.769119641423458e-06,
1.258204792975448e-05,
-5.148779109731549e-06,
7.967607729142401e-08,
1.1473611039036768e-06,
1.0064576599688735e-05,
4.101892955077346e-06,
-4.880315600530594e-07,
-1.0309180652257055e-05,
-9.864043022389524e-06,
2.749244458755129e-06,
-4.858302418142557e-06,
8.289614470413653e-07,
-4.286684998078272e-06,
-1.5943935068207793e-05,
5.677327408193378e-06,
-1.2359545507933944e-05,
-8.157444426615257e-06,
1.3629113709612284e-05,
```

```
    -5.655075710819801e-06,
    -1.2639788110391237e-05,
    -1.2539542694867123e-05,
    -3.008749672517297e-06,
    3.1721285722596804e-06,
    1.1716210792656057e-05,
    -1.3226584769654437e-06,
    -9.366031918034423e-06,
    -9.600781595509034e-06,
    -1.0696023309719749e-05,
    2.341779008929734e-06,
    -4.639327926270198e-06,
    -5.521186267287703e-06,
    -5.350485707822372e-07,
    -2.8162544367660303e-06,
    -1.3174344530852977e-05,
    0.47664734721183777,
};
const float taps_Q[NO_TAPS] = {
    2.3112465896701906e-06,
    -2.5339913918287493e-06,
    1.0334847502235789e-06,
    -2.5600795652280794e-06,
    7.035644671304908e-07,
    -1.7297938939009327e-06,
    1.9505830550770042e-06,
    6.712903086736333e-07,
    -1.4115727253738442e-06,
    -1.583596144882904e-06,
    9.955589348464855e-07,
    6.35011019767262e-06,
    8.391477877012221e-07,
    -2.160541043849662e-06,
    3.7129109387024073e-07,
    2.7778358457908325e-07,
    -8.538819997738756e-07,
    -2.310854824827402e-06,
    1.598152493897942e-06,
    -5.803669864690164e-06,
    7.139910280784534e-07,
    -5.911349944653921e-06,
    -1.3839721759723034e-06,
    -3.860774540953571e-06,
    1.6545880043850048e-07,
    6.258346729737241e-06,
    -8.755914677749388e-07,
    2.7600430030361167e-07,
    -4.070197519467911e-06,
```

3.7767097182950238e-06,
3.9047634459166147e-07,
1.173050691249955e-06,
-9.076916285266634e-07,
-2.7064450023317477e-06,
3.7631814393535024e-06,
-2.868160606794845e-07,
-2.5558745164744323e-06,
1.480780383644742e-06,
2.9067368956248174e-08,
3.7396112020360306e-06,
2.0375632630020846e-06,
-3.594490181058063e-07,
-4.2106607907044236e-06,
-4.088737114216201e-06,
2.449462726872298e-06,
3.60968527957084e-07,
4.419483502715593e-06,
6.074923931009835e-06,
-1.5101819599294686e-06,
-3.129682681901613e-07,
6.340423715300858e-05,
-6.010032848280389e-06,
7.286064942491066e-07,
2.2841581994725857e-06,
3.7413849440781632e-06,
-7.715139872743748e-06,
2.776697556328145e-06,
3.942078819818562e-06,
1.2662668495977414e-06,
4.418762273417087e-06,
-3.379535939984635e-07,
-1.268497612727515e-06,
2.3119307570595993e-06,
-1.8751863990473794e-06,
-4.812661245523486e-06,
2.385717152719735e-06,
-1.6294604847644223e-06,
4.965705556969624e-06,
-1.3544000694309943e-06,
5.656450525748369e-07,
-5.100544512970373e-06,
-6.396273420250509e-06,
-8.3853883552365e-06,
1.49812353811285e-06,
5.689849785994738e-06,
8.004937512851029e-07,
-1.1089219924542704e-06,

```
    -2.3092414380698756e-07,
    6.306718660198385e-06,
    8.916983460949268e-06,
    6.254105642256036e-07,
    6.286786174314329e-06,
    -5.596071787294932e-06,
    -3.3299087931482063e-07,
    0.0001227069296874106,
    -3.028003220606479e-06,
    -6.955522735552222e-07,
    7.611559340148233e-06,
    -1.5172694247667096e-06,
    -2.4282412596221548e-06,
    -6.8519602791639045e-06,
    3.1564602522848872e-06,
    -3.974055744038196e-06,
    -2.617355221445905e-06,
    -3.553588783233863e-07,
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    1.3415287867246661e-06,
    -2.727273113123374e-06,
    3.631369509093929e-06,
    0.4412938058376312,
};
const float weights_1[NEURONS_1*2] = {
    0.29674261808395386,
    -0.0006015824037604034,
    -0.3247993290424347,
    -0.10179246217012405,
    -0.5540593266487122,
    -0.15880891680717468,
    -0.43145641684532166,
    0.15039506554603577,
    -0.15197107195854187,
    -0.286664754152298,
    0.05332145839929581,
    0.38090306520462036,
    -0.12191034108400345,
    -0.3740667998790741,
    0.33779284358024597,
    0.07330404222011566,
    -0.03323235362768173,
    -0.21328257024288177,
    -0.4601798951625824,
    -0.2336806058883667,
    0.1883128136396408,
    0.19990867376327515,
    -0.05436862260103226,
```

```
    0.15885280072689056,
    -0.20350588858127594,
    0.3661174774169922,
    0.11024817824363708,
    -0.016073742881417274,
    -0.08577391505241394,
    -0.05652052164077759,
    -0.4811912178993225,
    0.08940330892801285,
    0.23470935225486755,
    -0.281951904296875,
    0.159349262714386,
    -0.22507931292057037,
    -0.26998093724250793,
    0.3192582428455353,
    -0.3553353548049927,
    0.22703561186790466,
};
const float biases_1[NEURONS_1] = {
    -0.08001810312271118,
    -0.006829914171248674,
    -0.0013002014020457864,
    -0.04833029955625534,
    0.0024987568613141775,
    0.03783488646149635,
    -0.056323397904634476,
    0.06757865846157074,
    0.007625691127032041 5,
    0.0013128650607541203,
    0.008932529017329216,
    -0.016176868230104446,
    0.003464589361101389,
    0.00788134802132845,
    -0.0012280507944524288,
    0.002717318246141076,
    -0.005990686360746622,
    -0.02280297502875328,
    0.0016060357447713614,
    0.04912131652235985,
};
const float weights_2[NEURONS_2*NEURONS_1] = {
    0.3642151355743408,
    0.2840445339679718,
    -0.36771360039711,
    -0.3322516083717346,
    0.023489054292440414,
    -0.1060083657503128,
    -0.23207281529903412,
```

-0.2322482466697693,
0.017104212194681168,
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-0.2865190804004669,
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    -0.44034722447395325,
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    -0.13343603909015656,
    0.3082911968231201,
    -0.0830802246928215,
    -0.19654437899589539,
};
const float biases_2[NEURONS_2] = {
    -0.021530786529183388,
    0.0018347740406170487,
    0.012500920332968235,
    0.011760037392377853,
    0.05706418678164482,
    0.01755109801888466,
    0.01940947026014328,
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    -0.005482502281665802,
    0.0011569521157070994,
    -0.005670054815709591,
    0.008498773910105228,
    0.0064272950403392315,
    -0.03052682615816593,
    -0.0023480223026126623,
    -0.01081751100718975,
};
const float weights_3[NEURONS_3*NEURONS_2] = {
    0.23057788610458374,
    0.01416162122040987,
    0.06018603965640068,
    0.008434011600911617,
    0.22406335175037384,
    -0.30079060792922974,
    -0.4784975051879883,
    0.015935493633151054,
    -0.04231182858347893,
    0.18219058215618134,
    0.06165073812007904,
    0.04439663514494896,
    0.33569133281707764,
    -0.14142458140850067,
    -0.40303924679756165,
```

```
    -0.3110697865486145,
    -0.20989853143692017,
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    0.04134777933359146,
    0.3609423339366913,
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    -0.21619077026844025,
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    -0.15146072208881378,
    -0.00040448168887471825,
    -0.18167459964752197,
    -0.09232154488563538,
    -0.0395924374461174,
};
const float biases_3[NEURONS_3] = {
    0.023129133507609367,
    -0.001444934168830514,
};
```

[32]:
```
paramSymb = intialise_paramSymb(M, nBits, seed=333)
symbTx = symbolSource(paramSymb)

slice = preprocess(symbTx)[0]

y_from_sw = dpd_model.predict(slice.reshape(1, 5000, 2), verbose=0)
print(np.concatenate((y_from_sw[0][:,0], y_from_sw[0][:,1]), axis=0))
```

```
[0.31582093 0.31581205 0.3192988  … 0.94745374 0.9444566  0.31748033]
```

[35]:
```
x_to_hw = np.concatenate((slice[:,0], slice[:,1]), axis=0)

print("const float x[10000] = {")
for val in x_to_hw:    print(f"    {val},")
print("};")
```

```
const float x[10000] = {
    0.31622776601683794,
```

```
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[55]:

```python
import tensorflow.keras.backend as K

sec_a_model = tf.keras.Model(inputs=dpd_model.input, outputs=dpd_model.
 ↪layers[1].output)
sec_a_out = sec_a_model.predict(slice.reshape(1, 5000, 2), verbose=0)
sec_a_out[0].reshape(-1)
```

[55]:

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array([ 0.31622776,  0.31622776,  0.31622776, …,  0.9486833 ,
       -0.9486833 , -0.9486833 ], dtype=float32)
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